FINAL PROJECT REPORT

International Mobile Banking UX Dataset Automation

TASK 07 – Team-Based Automation Project Development

Item	Details
Team size	2 members
Team member 1	Kanmani Suresh (DIT ID: DIT2026S1048)
Team member 2	Salman (DIT ID: DIT2026S1046)
Organisation	DefineInfoTech Pvt. Ltd. (DIT) – Internship 2026
Date	1 November 2026

1. Introduction

Mobile banking apps differ widely between countries. Data about their screens, features, security and usability is usually collected by hand in spreadsheets, so different contributors record the same information in different ways. Merging these files by hand is slow and error-prone.

This project automates the process. A Python pipeline loads the contributors' files, cleans and standardises them, validates every record against business rules, scores the UX quality of each screen, and produces a trusted master dataset and reports.

2. Objective and Business Outcome

Objective: build a working, tested automation that turns inconsistent contributor files into one clean, analysis-ready dataset.

Expected business outcome:

- One clean master dataset of banking-app screens with UX scores and grades.
- A separate list of rejected records, each with the reason, so contributors can correct them.
- Summaries comparing banks, regions and screen categories, with charts.
- Less manual effort, fewer data errors, and a repeatable process.

3. Tools and AI Usage

Tools: Python 3, pandas, openpyxl, matplotlib, pytest and Visual Studio Code.

AI usage: an AI assistant (Claude) was used for requirements analysis, architecture, synthetic dataset design, module-by-module code generation, test design and documentation. The generated code was run, tested, reviewed and modified by the team before submission. Important prompts and AI-assisted changes are recorded in AI_USAGE_LOG.md. No confidential or personal data was given to the AI tool.

Write 2–3 lines here in your own words about what the team reviewed and changed (for example: the grade limit change from 85 to 80 in config.py).

4. System Design

The solution is a five-stage pipeline. Each stage is a separate Python module, which keeps the code easy to understand and test.

Stage	Module	Purpose
1. Load	loader.py	Reads all CSV/Excel files, checks required columns, combines the rows
2. Clean	cleaner.py	Trims text and standardises countries, yes/no values, lists and dates
3. Validate	validator.py	Applies business rules; separates valid and rejected rows
4. Score	rules.py	Calculates the UX score (0–100) and grade

Stage	Module	Purpose
5. Report	report.py	Builds summaries, charts, the Excel workbook and a written summary

Supporting files: config.py (all settings and allowed values), pipeline.py (runs the stages in order), main.py (entry point with logging) and generate_dataset.py (creates the synthetic input data).

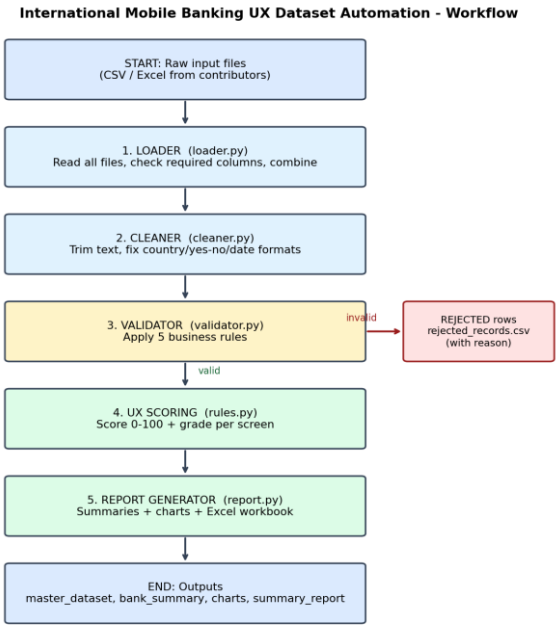


Figure 1: Workflow flowchart

5. Dataset

All data is synthetic. It covers 12 fictional banks in 12 countries, with 16 screens each, split across three contributor files (two CSV and one Excel). The columns are: record_id, bank_name, country, app_name, screen_no, screen_name, screen_category, features_present, auth_method, languages_supported, accessibility_flag, ux_rating, collected_by and collected_date.

To prove the automation works, the data deliberately contains 12 messy-but-fixable records (extra spaces, “USA”, “y”, mixed date formats), 8 invalid records and 2 duplicates.

6. Business Rules

ID	Rule	On failure
BR1	Mandatory fields must not be blank or unreadable	Reject
BR2	Country, category, authentication method and features must come from approved lists	Reject
BR3	ux_rating between 1 and 5; screen_no a positive whole number	Reject
BR4	collected_date cannot be in the future	Reject
BR5	(bank, app, screen_no) must be unique; first kept	Reject duplicate
BR6	Country aliases and Y/N values standardised	Auto-fix

ID	Rule	On failure
BR7	UX score = rating (60) + features (20) + accessibility (10) + languages (10)	Computed
BR8	Grade: ≥85 Excellent, ≥70 Good, ≥55 Fair, otherwise Poor	Computed

7. Testing

Fourteen automated test cases were written with pytest. All 14 passed.

ID	Test case	Expected result
TC01	Valid record	Accepted
TC02	Country alias “USA”	United States, North America
TC03	Blank bank_name	Rejected
TC04	Rating = 7	Rejected (out of range)
TC05	Rating = “abc”	Rejected
TC06	Category = “Gaming”	Rejected
TC07	Date in the future	Rejected
TC08	Duplicate record	First kept, second rejected
TC09	“y” and “N”	Yes and No
TC10	Unknown feature	Rejected
TC11	Maximum inputs	Score 100, Excellent
TC12	Minimum inputs	Score 12, Poor
TC13	Missing folder / bad file	FileNotFoundError / ValueError
TC14	End to end, 3 files	194 loaded, 184 valid, 10 rejected

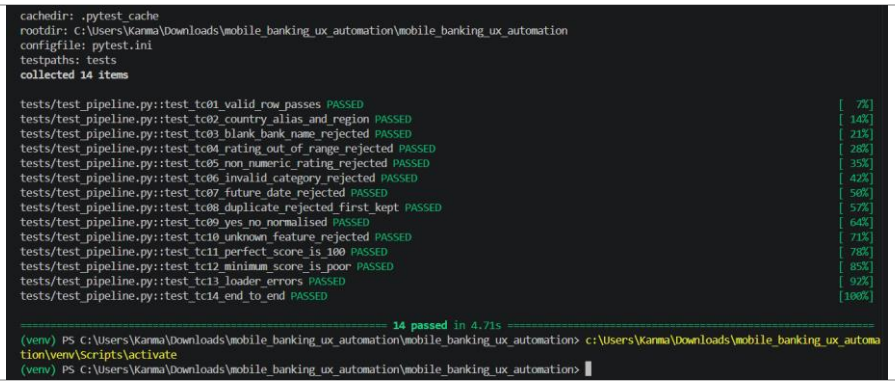


Figure 2: Test results

8. Results

Measure	Result
Files read	3
Records loaded	194
Valid records	184
Rejected records	10 (8 invalid + 2 duplicates)
Banks covered	12



Figure 3: Pipeline run summary

8.1 Why records were rejected

Reason	Records
bank name is missing or invalid	1
invalid screen_category	1
UX rating out of range 1-5	1
UX rating is missing or invalid	1
duplicate of an earlier record (bank, app, screen_no)	2
collected_date is in the future	1
unsupported country	1
invalid auth_method	1
unknown feature	1

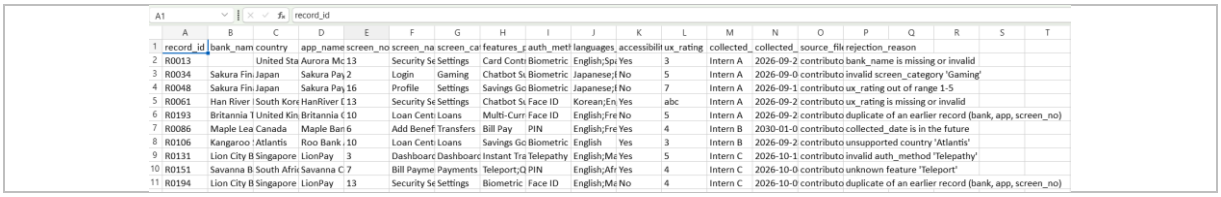


Figure 4: Rejected records with reasons

8.2 Bank ranking

Rank	Bank	Country	Screens	Avg rating	Avg score
1	Lion City Bank	Singapore	15	4.53	79.9
2	Aurora Bank	United States	15	4.33	74.9
3	Sakura Financial	Japan	14	4.36	73.6
4	Ganga Bank	India	16	3.81	73.0
5	Maple Leaf Bank	Canada	15	3.67	70.5
6	Han River Bank	South Korea	15	4.20	70.1
7	Britannia Trust	United Kingdom	16	4.00	68.5
8	Savanna Bank	South Africa	15	3.33	67.1
9	Emirates Crown Bank	United Arab Emirates	16	3.56	66.3
10	Rhein Bank	Germany	16	3.19	66.0
11	Kangaroo Savings	Australia	15	3.33	61.3
12	Verde Brasil Bank	Brazil	16	3.12	59.1

In this synthetic dataset, Lion City Bank (Singapore) has the highest average UX score (79.9) and Verde Brasil Bank (Brazil) the lowest (59.1).

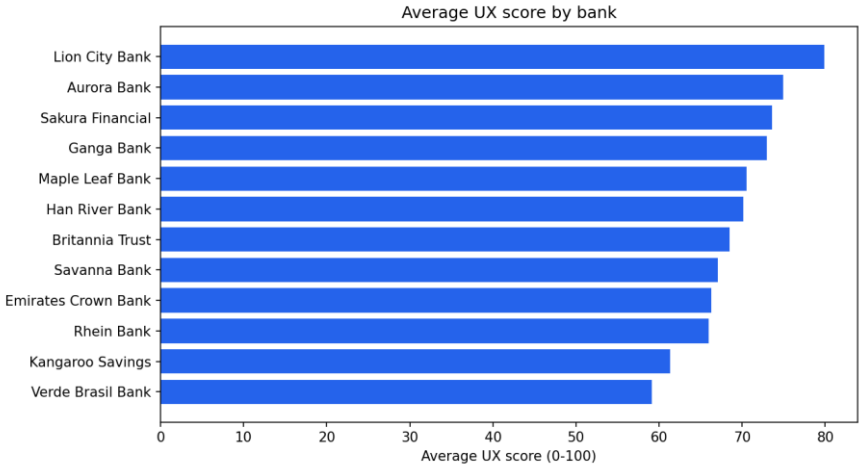


Figure 5: Average UX score by bank

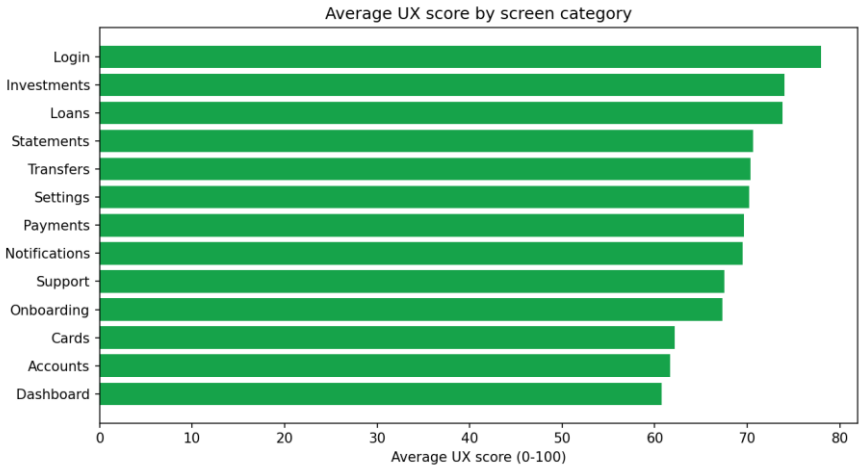
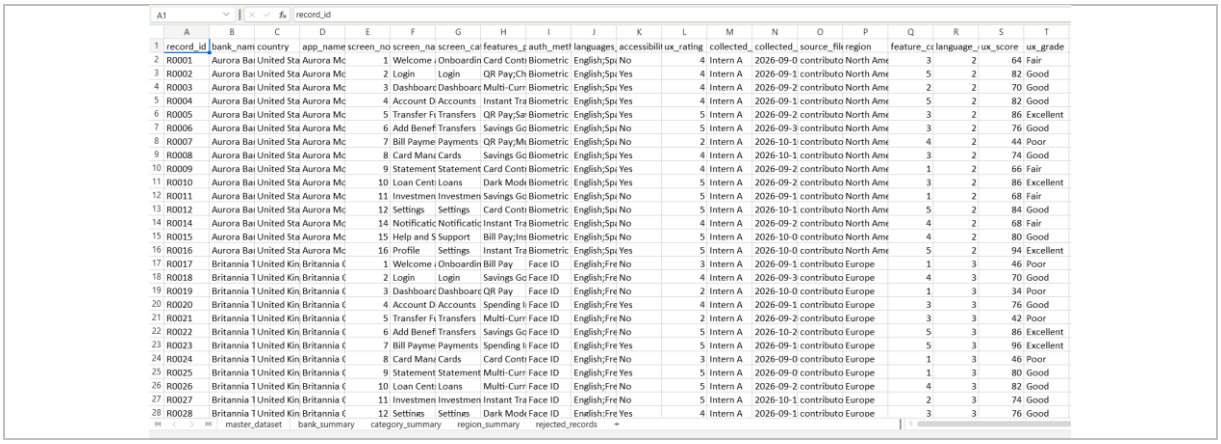
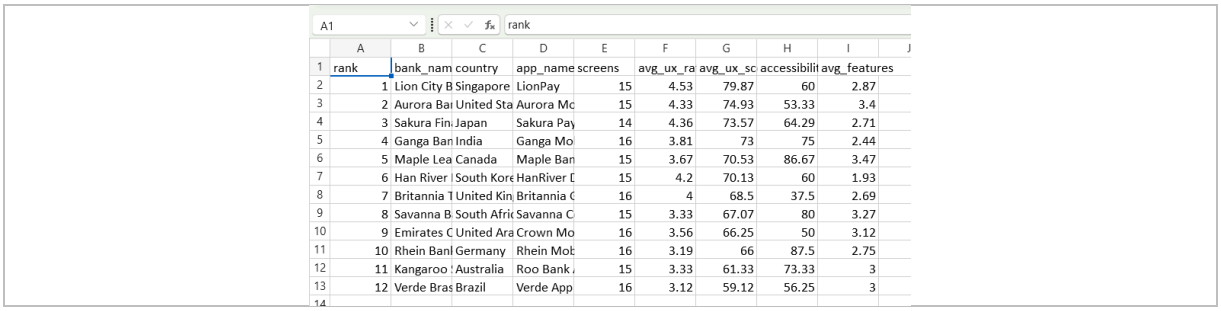


Figure 6: Average UX score by screen category (Login scores highest, Dashboard lowest in this dataset)



record_id	bank_nam	country	app_name	screen_no	screen_na	screen_ca	features_	auth_meth	languages	accessibili	ux_rating	collected_	collected_	source_fil	region	feature_co	language_	ux_score	ux_grade
R0001	Aurora Ban	United Sta	Aurora Mo	1	Welcome a	Onboardin	Card Conti	Biometric	English;Spa	No	4	Intern A	2026-09-0	contributo	North Ame	3	2	64	Fair
R0002	Aurora Ban	United Sta	Aurora Mo	2	Login	Login	QR Pay;Ch	Biometric	English;Spa	Yes	4	Intern A	2026-09-1	contributo	North Ame	5	2	82	Good
R0003	Aurora Ban	United Sta	Aurora Mo	3	Dashboarc	Dashboarc	Multi-Curr	Biometric	English;Spa	Yes	4	Intern A	2026-09-2	contributo	North Ame	2	2	70	Good
R0004	Aurora Ban	United Sta	Aurora Mo	4	Account D	Accounts	Instant Tra	Biometric	English;Spa	Yes	4	Intern A	2026-09-1	contributo	North Ame	5	2	82	Good
R0005	Aurora Ban	United Sta	Aurora Mo	5	Transfer Fl	Transfers	QR Pay;Sa	Biometric	English;Spa	Yes	5	Intern A	2026-09-2	contributo	North Ame	3	2	86	Excellent
R0006	Aurora Ban	United Sta	Aurora Mo	6	Add Benef	Transfers	Savings Go	Biometric	English;Spa	No	5	Intern A	2026-09-3	contributo	North Ame	3	2	76	Good
R0007	Aurora Ban	United Sta	Aurora Mo	7	Bill Payme	Payments	QR Pay;Mu	Biometric	English;Spa	No	2	Intern A	2026-10-1	contributo	North Ame	4	2	44	Poor
R0008	Aurora Ban	United Sta	Aurora Mo	8	Card Mana	Cards	Savings Go	Biometric	English;Spa	Yes	4	Intern A	2026-10-1	contributo	North Ame	3	2	74	Good
R0009	Aurora Ban	United Sta	Aurora Mo	9	Statement	Statement	Card Conti	Biometric	English;Spa	Yes	4	Intern A	2026-09-2	contributo	North Ame	1	2	66	Fair
R0010	Aurora Ban	United Sta	Aurora Mo	10	Loan Centi	Loans	Dark Mode	Biometric	English;Spa	Yes	5	Intern A	2026-09-2	contributo	North Ame	3	2	86	Excellent
R0011	Aurora Ban	United Sta	Aurora Mo	11	Investmen	Investmen	Savings Go	Biometric	English;Spa	No	5	Intern A	2026-09-1	contributo	North Ame	1	2	68	Fair
R0012	Aurora Ban	United Sta	Aurora Mo	12	Settings	Settings	Card Conti	Biometric	English;Spa	No	5	Intern A	2026-10-1	contributo	North Ame	5	2	84	Good
R0014	Aurora Ban	United Sta	Aurora Mo	14	Notificatic	Notificatic	Instant Tra	Biometric	English;Spa	No	4	Intern A	2026-09-2	contributo	North Ame	4	2	68	Fair
R0015	Aurora Ban	United Sta	Aurora Mo	15	Help and S	Support	Bill Pay;Ins	Biometric	English;Spa	No	5	Intern A	2026-10-0	contributo	North Ame	4	2	80	Good
R0016	Aurora Ban	United Sta	Aurora Mo	16	Profile	Settings	Instant Tra	Biometric	English;Spa	Yes	5	Intern A	2026-10-0	contributo	North Ame	5	2	94	Excellent
R0017	Britannia T	United Kin	Britannia (	1	Welcome a	Onboardin	Bill Pay	Face ID	English;Fre	No	3	Intern A	2026-09-1	contributo	Europe	1	3	46	Poor
R0018	Britannia T	United Kin	Britannia (	2	Login	Login	Savings Go	Face ID	English;Fre	No	4	Intern A	2026-09-3	contributo	Europe	4	3	70	Good
R0019	Britannia T	United Kin	Britannia (	3	Dashboarc	Dashboarc	QR Pay	Face ID	English;Fre	No	2	Intern A	2026-10-0	contributo	Europe	1	3	34	Poor
R0020	Britannia T	United Kin	Britannia (	4	Account D	Accounts	Spending li	Face ID	English;Fre	Yes	4	Intern A	2026-09-1	contributo	Europe	3	3	76	Good
R0021	Britannia T	United Kin	Britannia (	5	Transfer Fl	Transfers	Multi-Curr	Face ID	English;Fre	No	2	Intern A	2026-09-2	contributo	Europe	3	3	42	Poor
R0022	Britannia T	United Kin	Britannia (	6	Add Benef	Transfers	Savings Go	Face ID	English;Fre	No	5	Intern A	2026-10-2	contributo	Europe	5	3	86	Excellent
R0023	Britannia T	United Kin	Britannia (	7	Bill Payme	Payments	Spending li	Face ID	English;Fre	Yes	5	Intern A	2026-09-1	contributo	Europe	5	3	96	Excellent
R0024	Britannia T	United Kin	Britannia (	8	Card Mana	Cards	Card Conti	Face ID	English;Fre	No	3	Intern A	2026-09-0	contributo	Europe	1	3	46	Poor
R0025	Britannia T	United Kin	Britannia (	9	Statement	Statement	Multi-Curr	Face ID	English;Fre	Yes	5	Intern A	2026-09-0	contributo	Europe	1	3	80	Good
R0026	Britannia T	United Kin	Britannia (	10	Loan Centi	Loans	Multi-Curr	Face ID	English;Fre	No	5	Intern A	2026-09-2	contributo	Europe	4	3	82	Good
R0027	Britannia T	United Kin	Britannia (	11	Investmen	Investmen	Instant Tra	Face ID	English;Fre	No	5	Intern A	2026-10-1	contributo	Europe	2	3	74	Good
R0028	Britannia T	United Kin	Britannia (	12	Settings	Settings	Dark Mode	Face ID	English;Fre	Yes	4	Intern A	2026-09-1	contributo	Europe	3	3	76	Good

Figure 7a: ux_dataset_report.xlsx – master_dataset sheet



rank	bank_nam	country	app_name	screens	avg_ux_ra	avg_ux_sc	accessibilit	avg_features
1	Lion City B	Singapore	LionPay	15	4.53	79.87	60	2.87
2	Aurora Ban	United Sta	Aurora Mo	15	4.33	74.93	53.33	3.4
3	Sakura Fin	Japan	Sakura Pay	14	4.36	73.57	64.29	2.71
4	Ganga Ban	India	Ganga Mo	16	3.81	73	75	2.44
5	Maple Lea	Canada	Maple Ban	15	3.67	70.53	86.67	3.47
6	Han River	South Kore	HanRiver E	15	4.2	70.13	60	1.93
7	Britannia T	United Kin	Britannia C	16	4	68.5	37.5	2.69
8	Savanna B	South Afric	Savanna C	15	3.33	67.07	80	3.27
9	Emirates C	United Ara	Crown Mo	16	3.56	66.25	50	3.12
10	Rhein Banl	Germany	Rhein Mob	16	3.19	66	87.5	2.75
11	Kangaroo	Australia	Roo Bank	15	3.33	61.33	73.33	3
12	Verde Bras	Brazil	Verde App	16	3.12	59.12	56.25	3

Figure 7b: ux_dataset_report.xlsx – bank_summary sheet

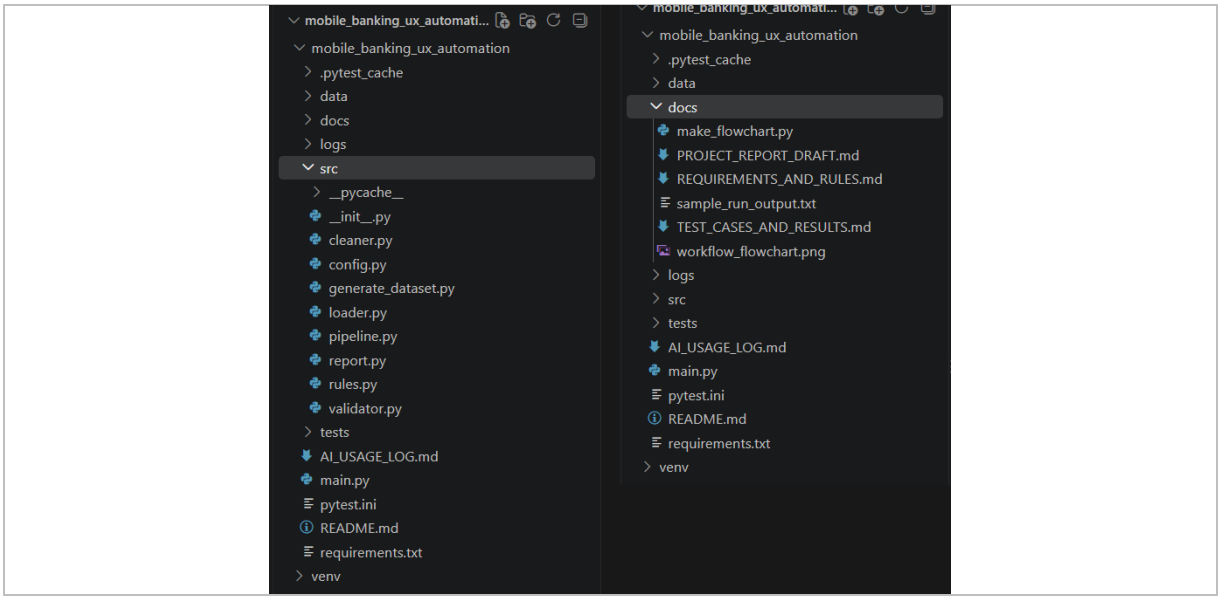


Figure 8a: Project structure – src folder (left) and docs folder (right)

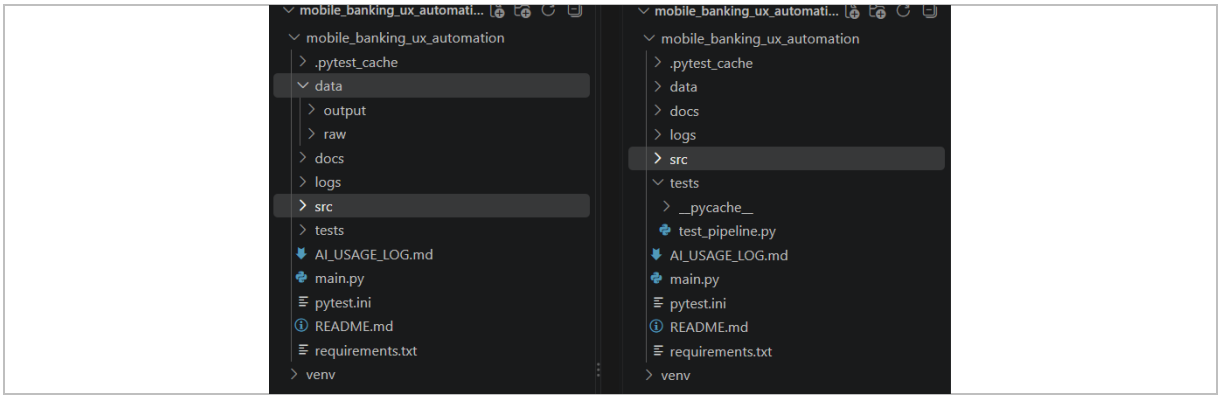


Figure 8b: Project structure – data folder (left) and tests folder (right)

9. Challenges and Learning

Write this section yourself in 3–4 sentences. Ideas: handling inconsistent formats; keeping the original values in the rejected file; why synthetic data was used; why AI-generated code must be reviewed and tested; what you learned about pandas or testing.

10. Limitations and Future Work

- Input must already be tabulated; screenshots of apps are not read automatically.
- Score weights and approved lists are assumptions and can be changed in config.py.

- Future work: OCR/image input, a dashboard, database storage and scheduled runs.

11. Conclusion

The project delivers a tested, modular Python automation that converts inconsistent contributor files into a validated, scored master dataset with clear reports. Every business rule is covered by automated tests, and every rejected record is explained.

12. Team Members and Contributions

Team member	DIT ID	Contribution
Kanmani Suresh	DIT2026S1048	(write her modules / tasks)
Salman	DIT2026S1046	(write his modules / tasks)

Fill in the Contribution column: DIT requires every team member to contribute. For example, who built, tested and documented which modules, and who prepared the report and presentation.

13. Project Deliverables

Deliverable	File / location
Source code	main.py and src/
Input dataset	data/raw/
Business rules and requirements	docs/REQUIREMENTS_AND_RULES.md
Flowchart	docs/workflow_flowchart.png
Test cases and results	tests/test_pipeline.py, docs/TEST_CASES_AND_RESULTS.md
Generated outputs	data/output/
README	README.md
AI usage record	AI_USAGE_LOG.md